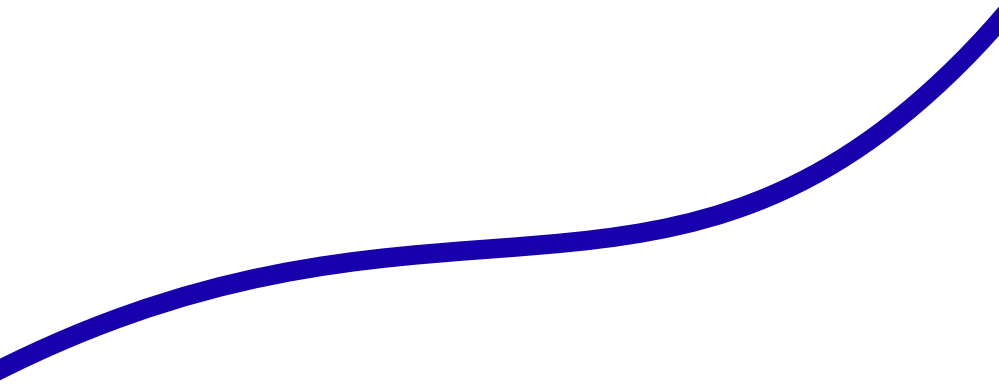
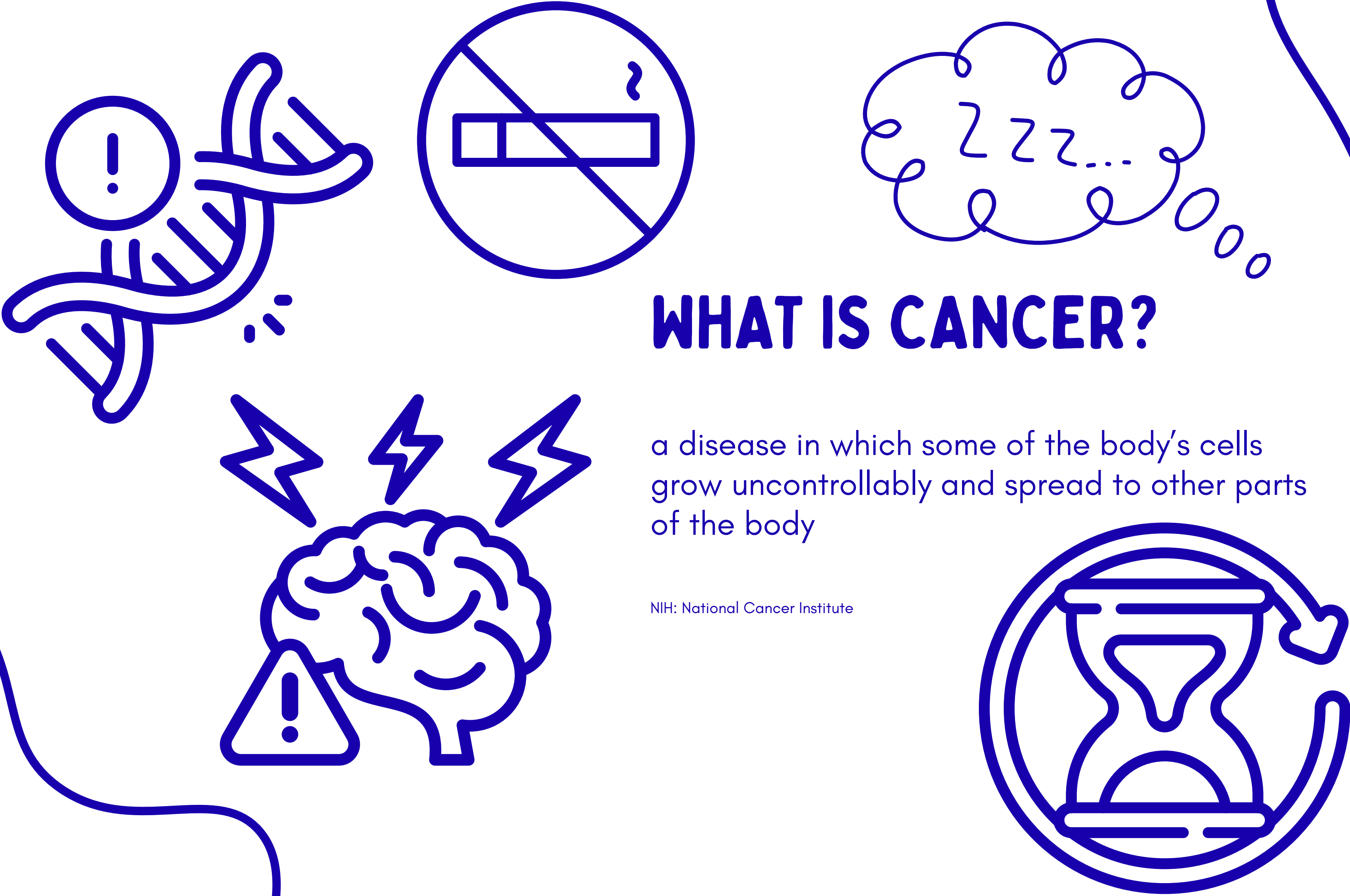




SOCIOECONOMIC FACTORS AND THEIR EFFECT ON CANCER MORTALITY RATES

Catherine Wu



WHAT IS CANCER?

a disease in which some of the body's cells grow uncontrollably and spread to other parts of the body

NIH: National Cancer Institute

WE DON'T LIVE IN A VACUUM

Socioeconomic factors have been shown to influence health outcomes.

Barakat & Konstantinidis (2023)



DATA

- Several Sources from 2010–2016:
 - 2013 United States Census
 - National Library of Medicine – Clinical Trials
 - National Cancer Institute
- n = 3047
- Response Variable (1)
 - Rate of Cancer Mortalities per 100,000 Residents of U.S. Counties
- Summary of Predictor Variables (29)
 - Cancer Incidence Rate
 - Level of Wealth
 - Education Level
 - Marital Status/Household Size
 - Insurance
 - Race

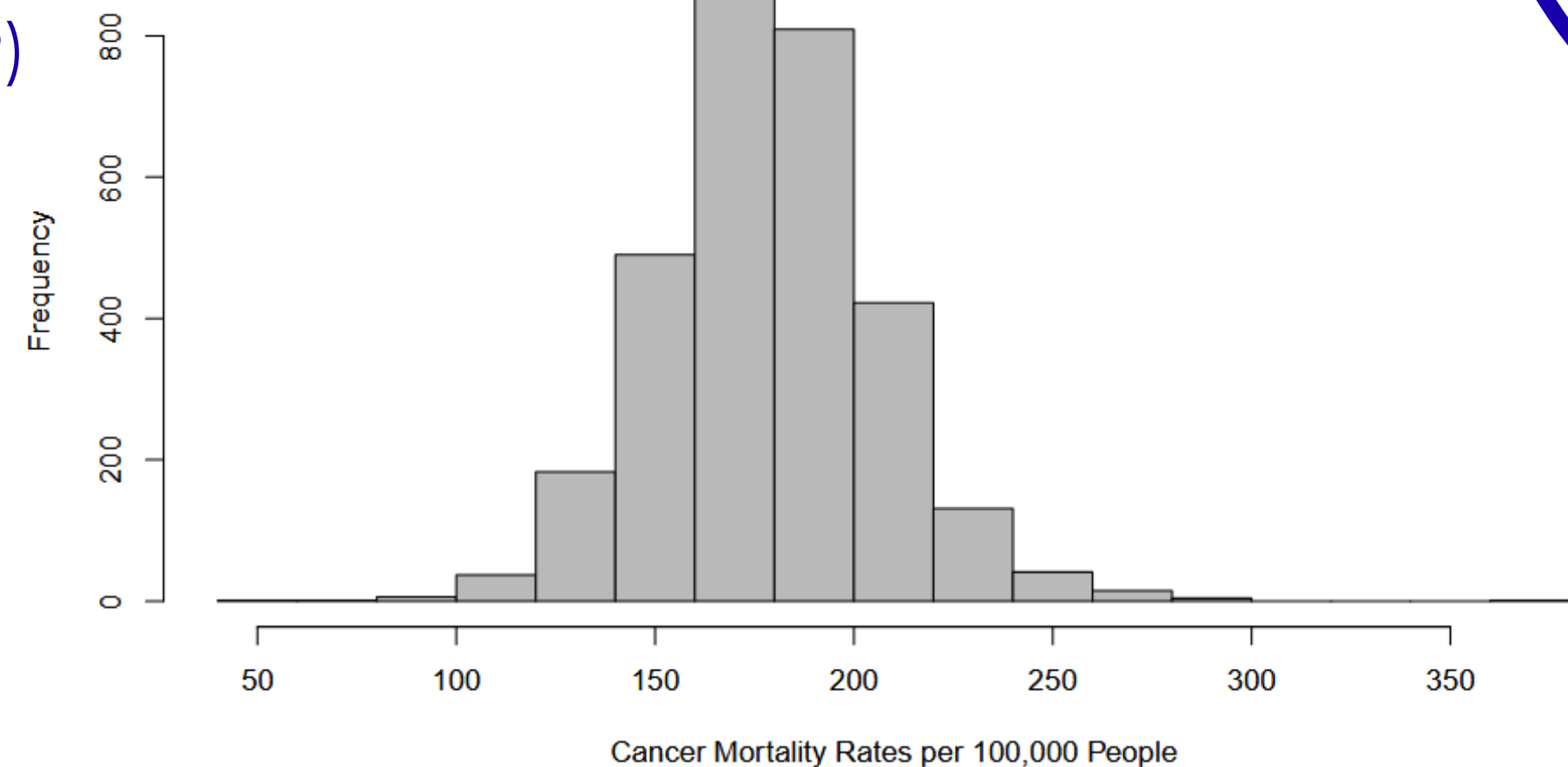


Figure 1: Histogram of Cancer Mortality Rates in U.S. Counties

EXPLORATORY DATA ANALYSIS

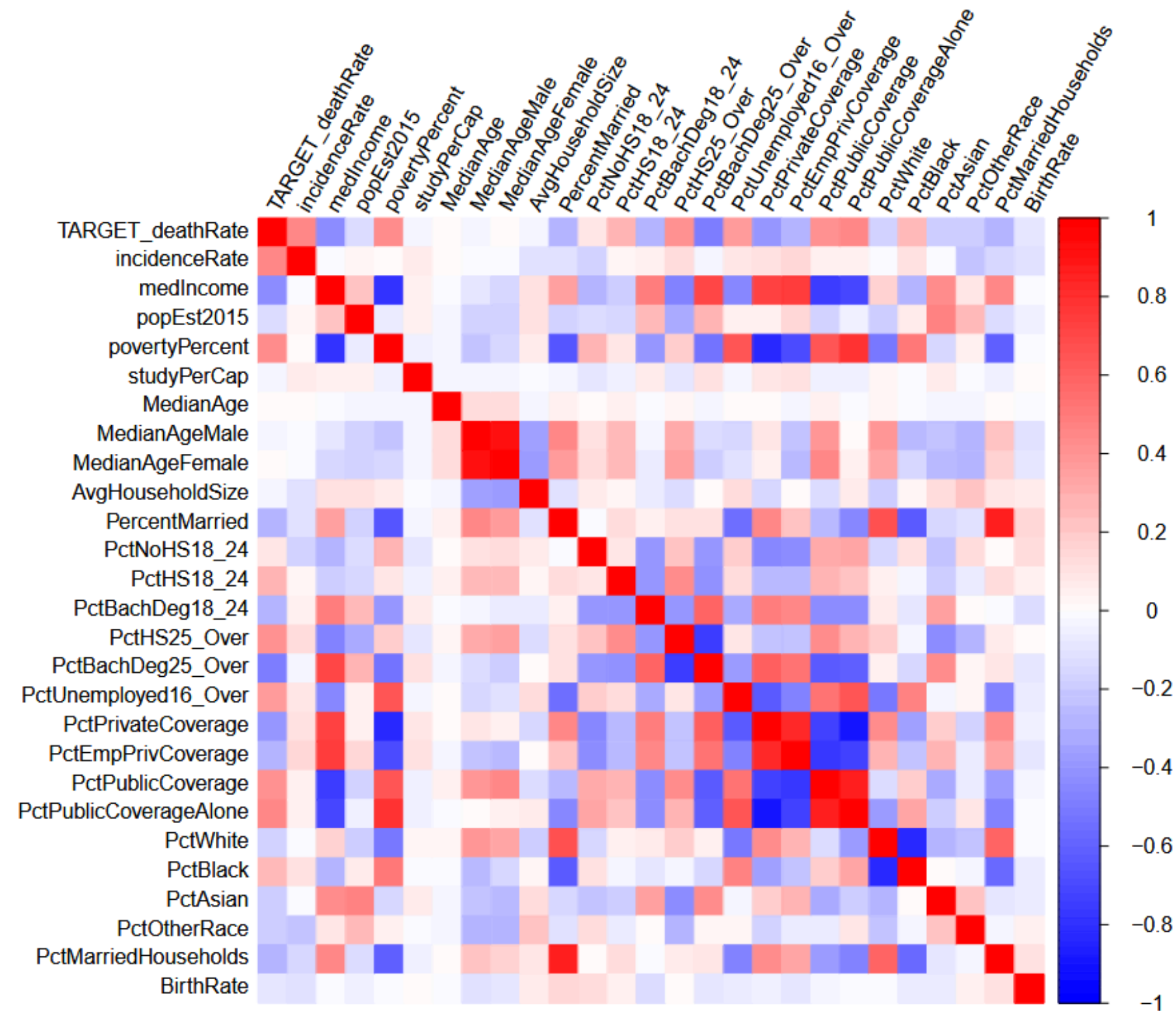


Figure 2: Correlation Heatmap of Socioeconomic Factors and Cancer Mortality

METHODS

- Multiple Linear Regression Model Assumptions
 - The predictor variables are fixed or conditioned upon
 - The response variable have a constant variance
 - The errors are normally distributed
 - The errors are independent
 - The parameters of the model have a linear relationship

$$Y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_k x_{ik} + \varepsilon_i$$

- Least Squares Estimation
- Goodness of Fit
- Model Diagnostics
- Statistical Inference
- R

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \cdots + \hat{\beta}_k x_k$$

$$r^2 = \frac{SSR}{SST} = 1 - \frac{SSE}{SST}$$

RESULTS OF THE INITIAL MODEL

Table 2: Summary of coefficients $\hat{\beta}$'s of the predictor variables and their significance.

	Estimate ($\hat{\beta}$)	Std. Error	t value	Pr(> t)
(Intercept)	124.369	13.235	9.397	1.1E-20***
Cancer Incidence Rate	0.192	0.007	26.657	8.9E-141***
Median Income	9.2E-05	8.1E-05	1.135	0.256
Population	-1.9E-06	1.3E-06	-1.462	0.144
Percent of Populace in Poverty	0.562	0.153	3.673	2.4E-04***
Number of Cancer-Related Clinical Trials per Capita	-2.1E-04	0.001	-0.310	0.757
Median Age	-0.003	0.008	-0.419	0.675
Median Age of Male Residents	-0.501	0.209	-2.395	0.017*
Median Age of Female Residents	0.057	0.215	0.265	0.791
Mean Household Size	0.475	0.963	0.493	0.622
Percent of Residents who are Married	0.812	0.147	5.526	3.5E-08***
Percent of Married Households	-0.879	0.140	-6.278	3.9E-10***
Birth Rate	-0.982	0.194	-5.069	4.2E-07***
Percent of County Residents Aged 18-24 Highest Education: less than High School	-0.133	0.056	-2.371	0.018*
Percent of County Residents Aged 18-24 Highest Education: High School Diploma	0.254	0.049	5.155	2.7E-07***
Percent of County Residents Aged 18-24 Highest Education: Bachelor's Degree	-0.080	0.109	-0.739	0.460
Percent of County Residents Over Age 25 Highest Education: High School Diploma	0.413	0.097	4.263	2.1E-05***
Percent of County Residents Over Age 25 Highest Education: Bachelor's Degree	-1.191	0.154	-7.733	1.4E-14***
Percent of Unemployed County Residents Over Age 16	0.451	0.161	2.812	0.005**
Percent of County Residents with Private Health Coverage	-0.543	0.131	-4.150	3.4E-05***
Percent of County Residents with Employee-Provided Private Health Coverage	0.337	0.103	3.265	0.001**
Percent of County Residents with Government-Provided Health Coverage	0.032	0.218	0.145	0.885
Percent of County Residents with Government-Provided Health Coverage Alone	0.068	0.276	0.248	0.804
Percent of County Residents who Identify as White	-0.096	0.057	-1.664	0.096 .
Percent of County Residents who Identify as Black	-0.025	0.056	-0.455	0.649
Percent of County Residents who Identify as Asian	-0.005	0.189	-0.029	0.977
Percent of County Residents who Identify as a Race Other than White, Black, or Asian	-0.911	0.125	-7.303	3.6E-13***

Where level of significance is indicated by: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Table 3: Initial Multiple Linear Regression Model Summary and ANOVA Test

Source	r^2	Adjusted r^2	Sum of Squares	d.f.	MSE	F-statistic	p-value
Regression	0.5136	0.5094	1204859	26	46340.7	122.65	<2.2E-16
Error			1141007	3020	377.8		
Total			2345866	3046			

MODEL DIAGNOSTICS OF THE INITIAL MODEL

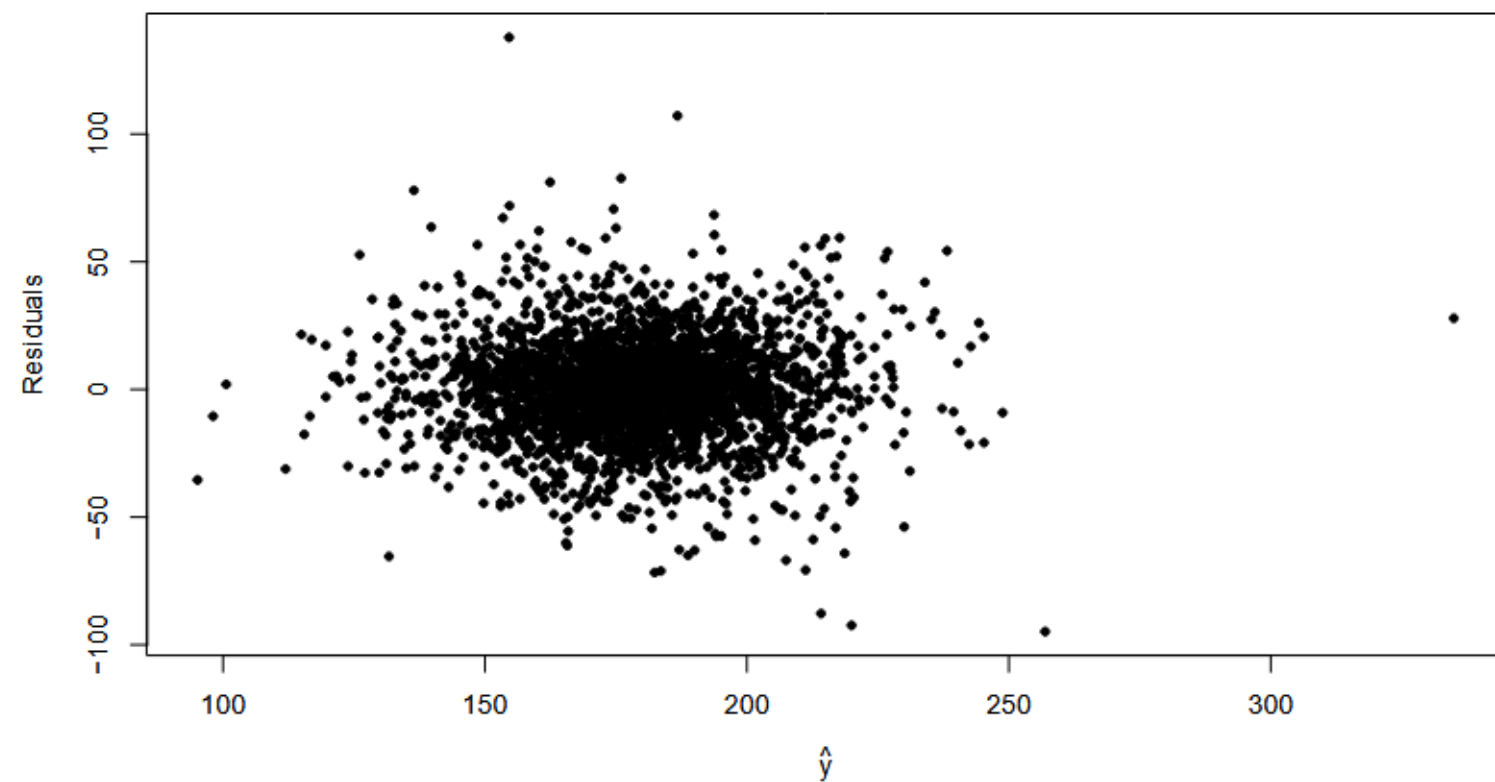


Figure 4: Scatter Plot of Residuals vs. Fitted Values ($\hat{y}$)

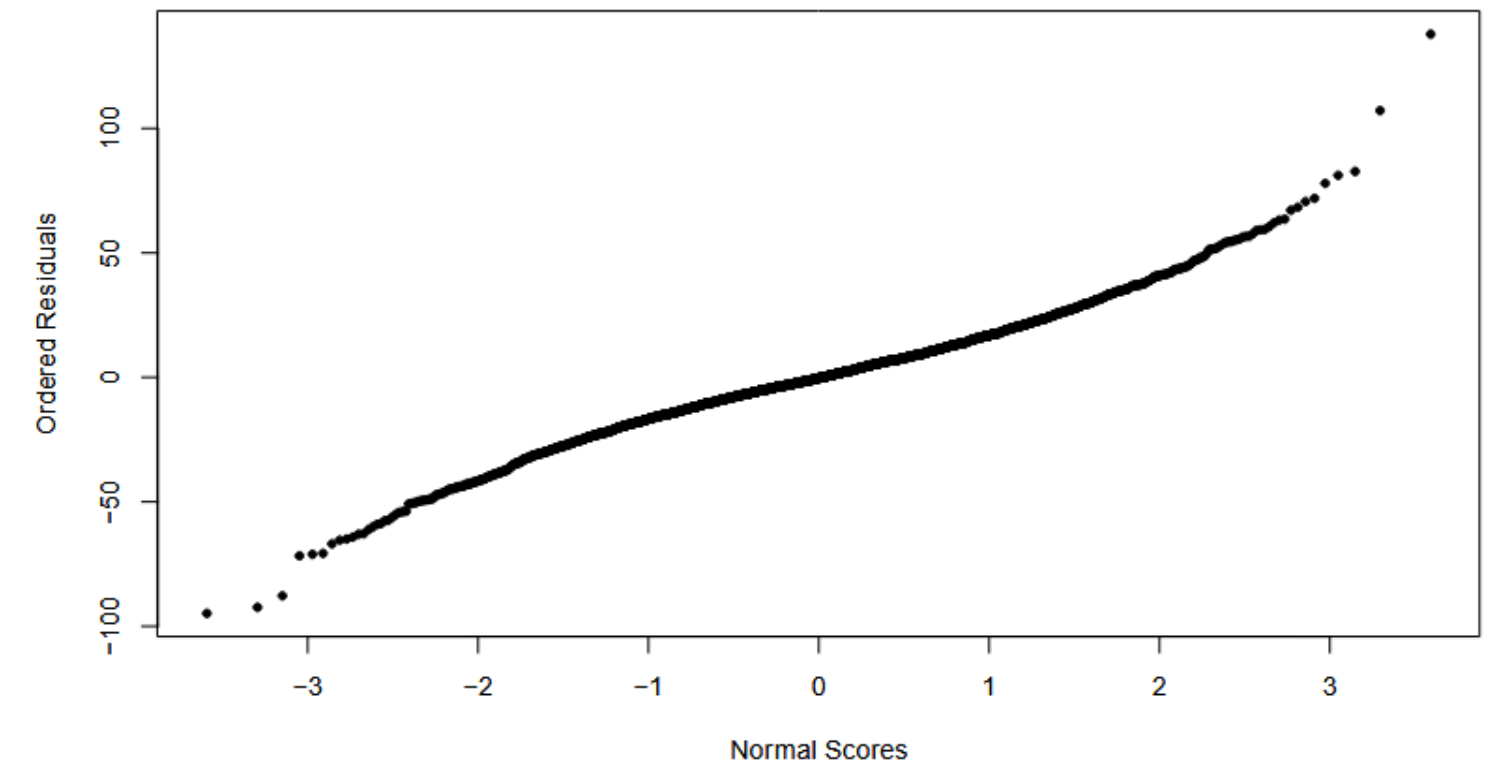


Figure 5: Normal Plot of Residuals

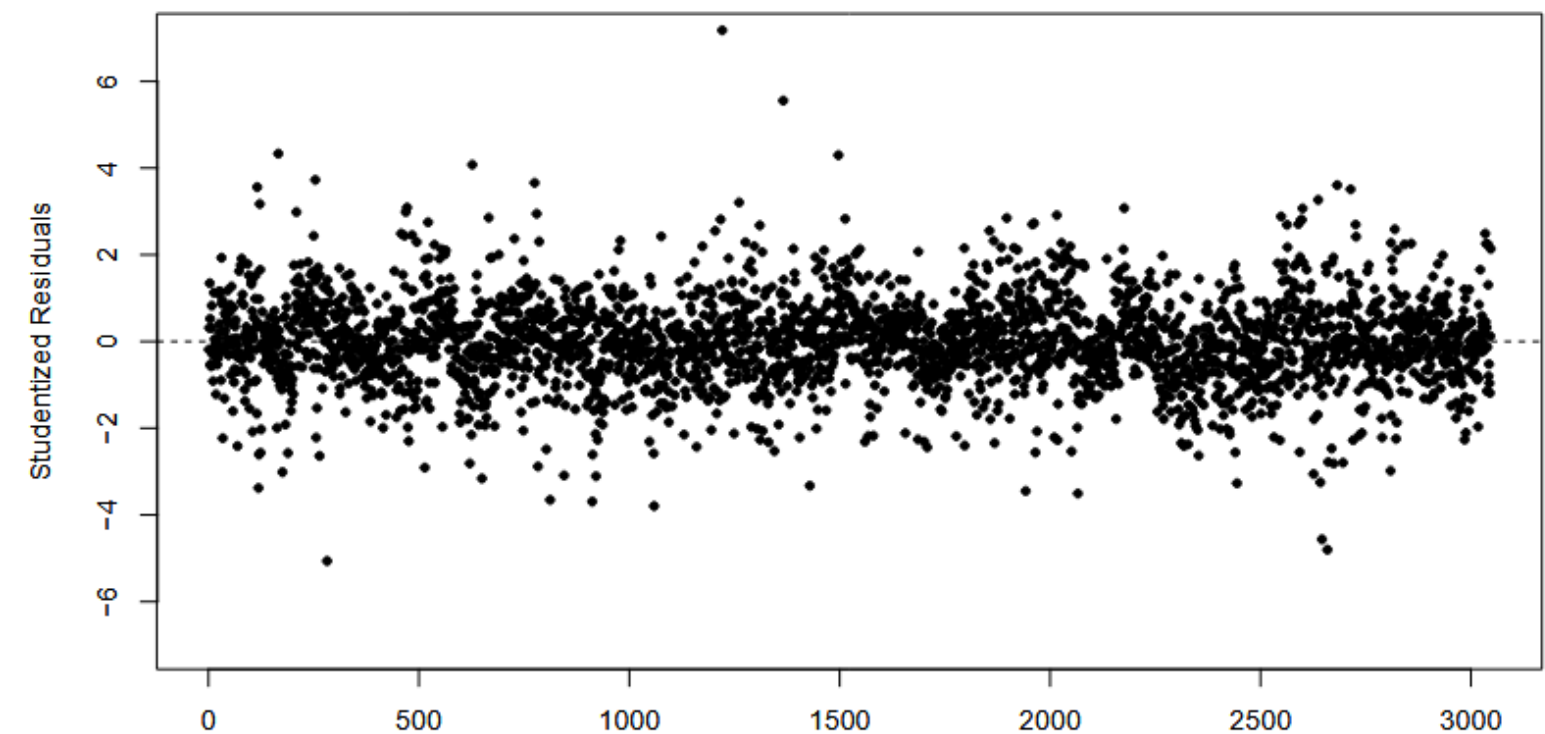


Figure 6: Scatter Plot of Studentized Residuals

FITTING THE MODEL - REMOVING OUTLIERS

- Influential observations
 - Leverage value $> 2(k + 1)/n$
- Outliers
 - $| \text{Studentized residual} | > 2$

Table 4: Multiple Linear Regression Model Summary with Outliers Removed and ANOVA Test Comparison with Initial Model

Source	r^2	Adjusted r^2	Sum of Squares	d.f.	MSE	F-statistic	p-value
Regression	0.6189	0.6151	948245	26	36471.0	162.71	$<2.2\text{E-}16$
Error			583894	2605	224.14		
Total			1532138	2631			

Table 5: Summary of coefficients ($\hat{\beta}$'s) of the predictor variables and their significance after removing outliers.

	Estimate ($\hat{\beta}$)	Std. Error	t value	Pr(> t)
(Intercept)	159.843	13.308	12.011	2.201E-32 ***
Cancer Incidence Rate	0.175	0.007	25.298	1.917E-126 ***
Median Income	0.000	0.000	1.872	0.061 .
Population	-0.000	0.000	-0.803	0.422
Percent of Populace in Poverty	0.632	0.147	4.313	1.670E-05 ***
Number of Cancer-Related Clinical Trials per Capita	-0.000	0.001	-0.514	0.608
Median Age of County Residents	0.535	1.054	0.507	0.612
Median Age of Male County Residents	-0.979	0.595	-1.645	0.100
Median Age of Female County Residents	-0.068	0.539	-0.126	0.900
Mean Household Size	-1.157	1.327	-0.872	0.383
Percent of Married Individuals	0.715	0.150	4.767	1.976E-06 ***
Percent of Married Households	-0.777	0.143	-5.446	5.637E-08 ***
Birth Rate	-0.971	0.186	-5.222	1.914E-07 ***
Percent of County Residents Aged 18-24 Highest Education: less than High School	-0.112	0.052	-2.165	0.030 *
Percent of County Residents Aged 18-24 Highest Education: High School Diploma	0.271	0.044	6.179	7.467E-10 ***
Percent of County Residents Aged 18-24 Highest Education: Bachelor's Degree	-0.033	0.108	-0.303	0.762
Percent of County Residents Over Age 25 Highest Education: High School Diploma	0.520	0.086	6.008	2.146E-09 ***
Percent of County Residents Over Age 25 Highest Education: Bachelor's Degree	-1.119	0.140	-7.982	2.141E-15 ***
Percent of Unemployed County Residents Over Age 16	0.830	0.149	5.572	2.774E-08 ***
Percent of County Residents with Private Health Coverage	-0.813	0.127	-6.424	1.570E-10 ***
Percent of County Residents with Employee-Provided Private Health Coverage	0.478	0.099	4.820	1.522E-06 ***
Percent of County Residents with Government-Provided Health Coverage	0.072	0.202	0.356	0.721
Percent of County Residents with Government-Provided Health Coverage Alone	-0.200	0.256	-0.780	0.435
Percent of County Residents who Identify as White	-0.282	0.076	-3.712	2.101E-04 ***
Percent of County Residents who Identify as Black	-0.210	0.075	-2.815	0.005 **
Percent of County Residents who Identify as Asian	0.111	0.296	0.374	0.708
Percent of County Residents who Identify as a Race Other than White, Black, or Asian	-1.765	0.161	-10.990	1.702E-27 ***

Where level of significance is indicated by: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

FITTING THE MODEL - MULTICOLLINEARITY

- Variance Inflation Factor (VIF)



incidenceRate	medIncome	popEst2015	povertyPercent	studyPerCap
1.277029	8.640512	1.938660	8.715075	1.121246
MedianAge	MedianAgeMale	MedianAgeFemale	AvgHouseholdSize	PercentMarried
308.876183	99.022188	82.548725	1.843226	10.586769
PctNoHS18_24	PctHS18_24	PctBachDeg18_24	PctHS25_Over	PctBachDeg25_Over
1.667364	1.684114	2.108779	4.046446	6.098626
PctUnemployed16_Over	PctPrivateCoverage	PctEmpPrivCoverage	PctPublicCoverage	PctPublicCoverageAlone
2.596118	18.300678	9.445012	27.111597	25.011565
PctWhite	PctBlack	PctAsian	PctOtherRace	PctMarriedHouseholds
14.622983	12.921711	2.610098	1.953138	9.026606
BirthRate				
1.196991				

Before	Median Age of County Residents	0.535	1.054	0.507	0.612
	Median Age of Male County Residents	-0.979	0.595	-1.645	0.100
	Median Age of Female County Residents	-0.068	0.539	-0.126	0.900
After	Median Age of Male County Residents	-0.697	0.212	-3.282	0.001 **
	Median Age of Female County Residents	0.183	0.213	0.859	0.390

Table 6: Multiple Linear Regression Model Summary with Outliers and Median Age of County Residents
Predictor Variable Removed and ANOVA Test Comparison with Linear Model with Outliers Removed

Source	r^2	Adjusted r^2	Sum of Squares	d.f.	MSE	F-statistic	p-value
Regression	0.6189	0.6152	948187	25	37939.5	169.26	<2.2E-16
Error			583951	2606	224.08		
Total			1532138	2631			

FITTING THE MODEL - MULTICOLLINEARITY

incidenceRate	medIncome	popEst2015	povertyPercent	studyPerCap
1.273798	8.635456	1.938242	8.666930	1.121176
MedianAgeMale	MedianAgeFemale	AvgHouseholdSize	PercentMarried	PctNoHS18_24
12.614598	12.915953	1.832977	10.067657	1.652816
PctHS18_24	PctBachDeg18_24	PctHS25_Over	PctBachDeg25_Over	PctUnemployed16_Over
1.682388	2.106749	4.040808	6.050412	2.569540
PctPrivateCoverage	PctEmpPrivCoverage	PctPublicCoverage	PctPublicCoverageAlone	PctWhite
18.211066	9.428404	27.102349	25.004078	14.618127
PctBlack	PctAsian	PctOtherRace	PctMarriedHouseholds	BirthRate
12.878388	2.610028	1.952217	8.775590	1.196974

Table 8: Multiple Linear Regression Model Summary with Outliers, Median Age of County Residents and Percentage of County Residents with Government-Provided Health Coverage Predictor Variables Removed and ANOVA Test Comparison with Linear Model with Outliers and Median Age of County Residents Predictor Variable Removed

Source	r^2	Adjusted r^2	Sum of Squares	d.f.	MSE	F-statistic	p-value
Regression	0.6188	0.6153	57.646	1	57.646	0.2572	0.6121
Error			583951	2605	224.2		
Total			583894	2606			

- Before

Percent of County Residents with Private Health Coverage	-0.818	0.126	-6.477	1.118E-10 ***
Percent of County Residents with Employee-Provided Private Health Coverage	0.480	0.099	4.846	1.335E-06 ***
Percent of County Residents with Government-Provided Health Coverage	0.074	0.202	0.366	0.715
Percent of County Residents with Government-Provided Health Coverage Alone	-0.202	0.256	-0.789	0.430

- After

	Estimate ($\hat{\beta}$)	Std. Error	t value	Pr(> t)
Percent of County Residents with Private Health Coverage	-0.795	0.111	-7.195	8.1E-13 ***
Percent of County Residents with Employee-Provided Private Health Coverage	0.463	0.087	5.307	1.2E-07 ***
Percent of County Residents with Government-Provided Health Coverage Alone	-0.123	0.135	-0.908	0.364

MODEL DIAGNOSTICS OF THE FITTED MODEL

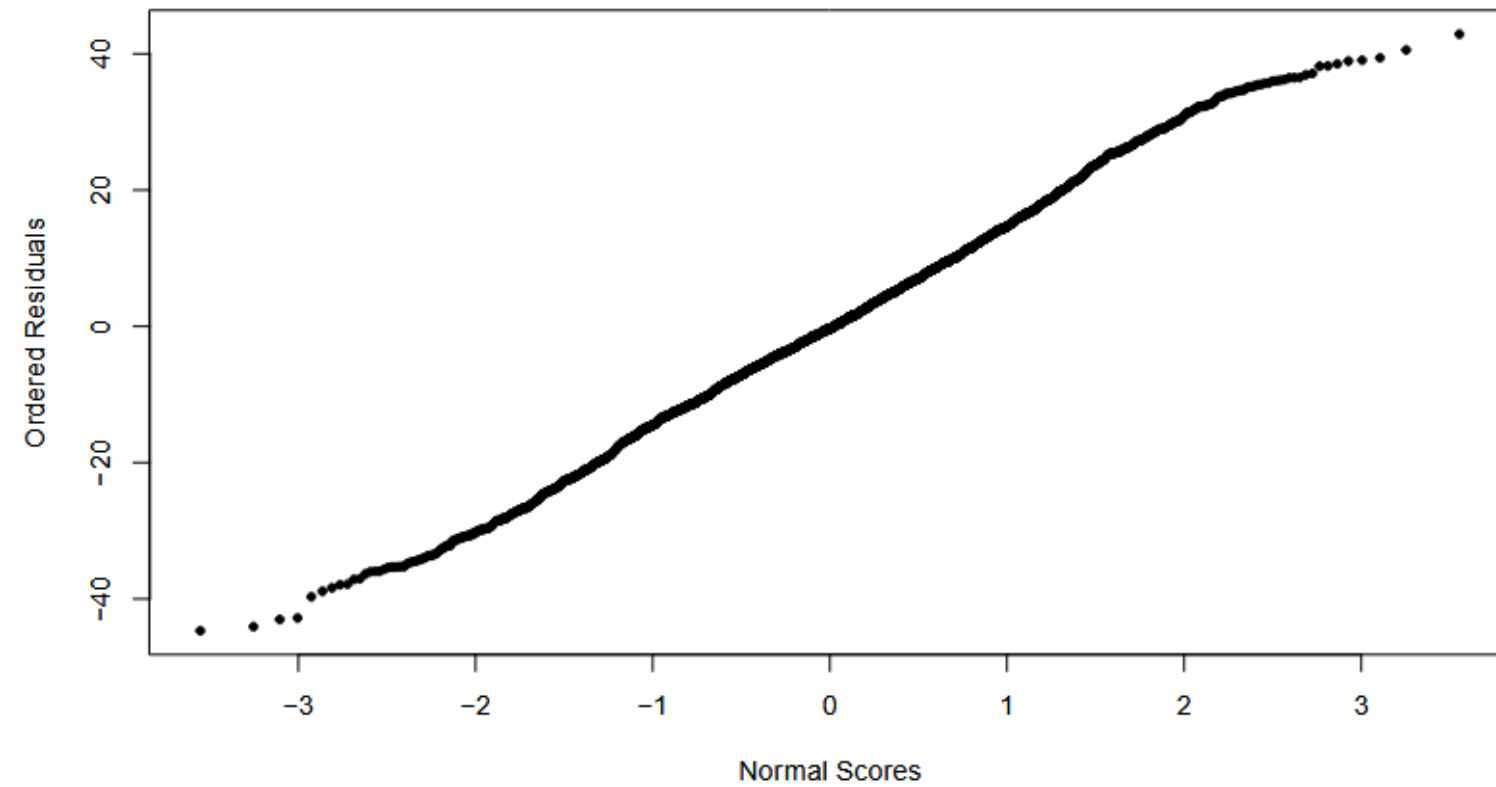


Figure 8: Normal Plot of Residuals

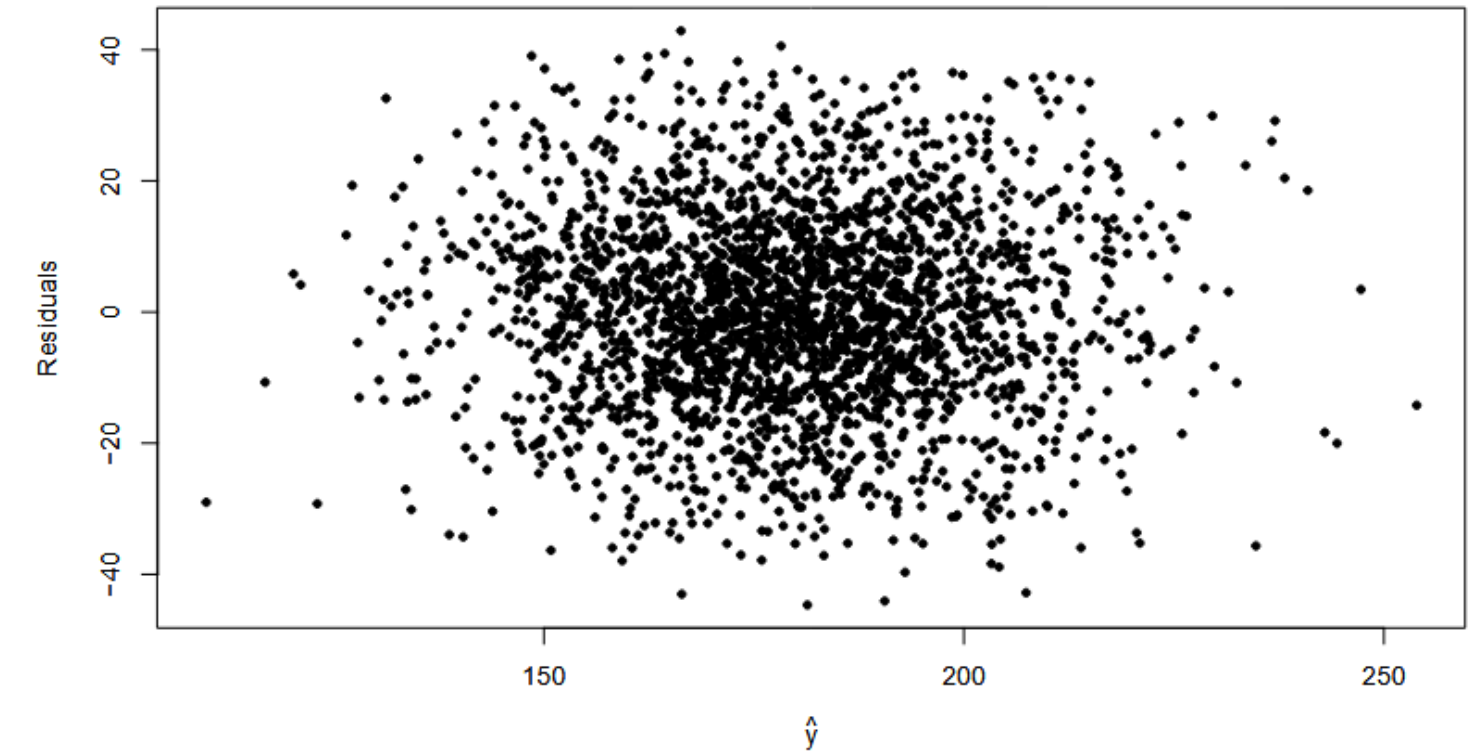


Figure 7: Scatter Plot of Residuals vs. Fitted Values ($\hat{y}$)

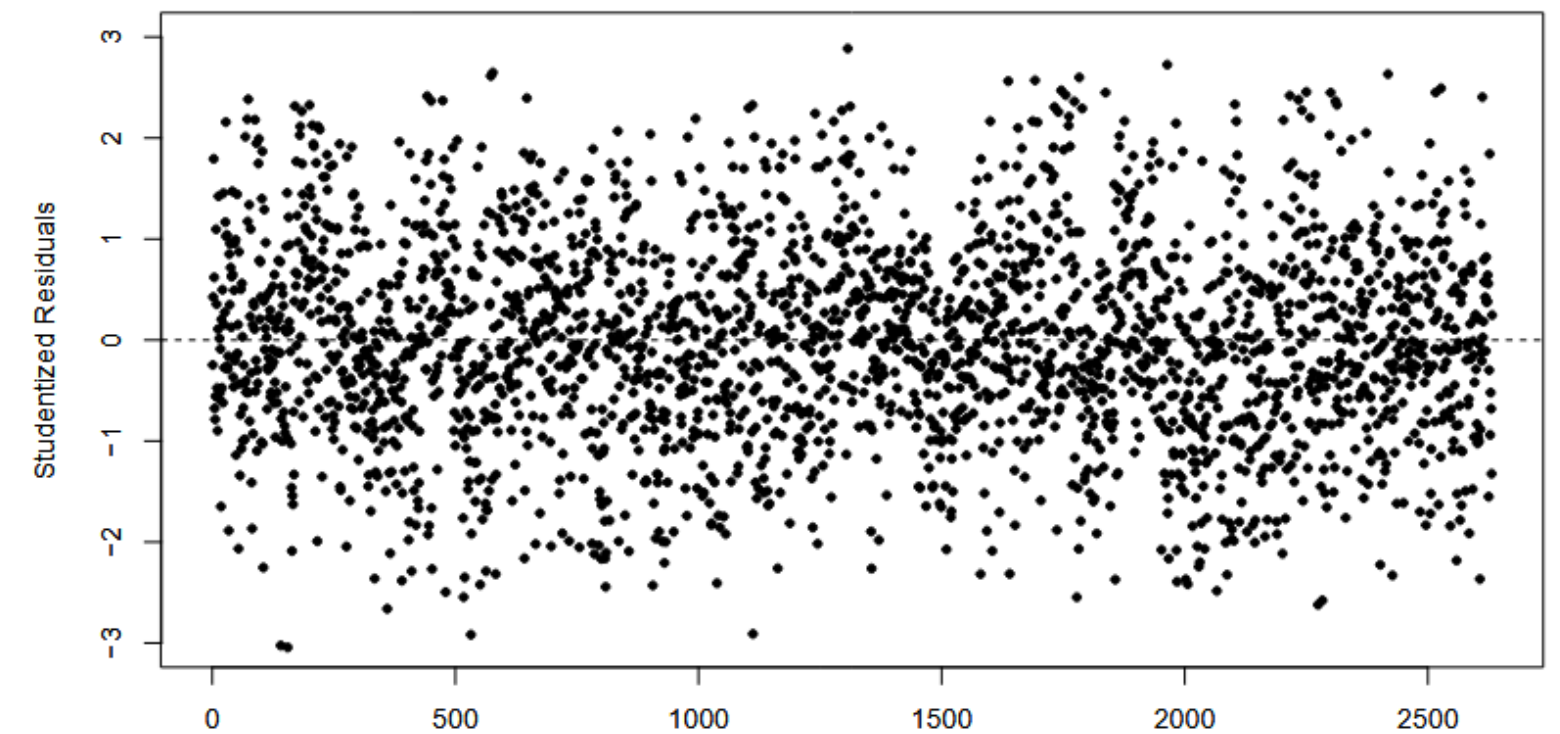


Figure 9: Scatter Plot of Studentized Residuals

STATISTICAL INFERENCE ON REGRESSION COEFFICIENTS

- Statistical Inference on individual predictor variables
 - $H_0: \beta_j = 0$; indicates the predictor variables is not related to y
 - $H_1: \beta_j \neq 0$; indicates the predictor variables is useful in predicting y , holding all other predictor variables constant
- Cancer Incidence Rate
- Percent of Populace in Poverty
- median Age of Male County Residents
- Percent of Married Individuals
- Percent of Married Households
- Birth Rate
- Percent of County Residents Aged 18–24 Highest Education: less than High School
- Percent of County Residents Aged 18–24 Highest Education: High School Diploma
- Percent of County Residents Over Age 25 Highest Education: High School Diploma
- Percent of County Residents Over Age 25 Highest Education: Bachelor's Degree
- Percent of Unemployed County Residents Over Age 16
- Percent of County Residents with Private Health Coverage
- Percent of County Residents with Employee-Provided Private Health Coverage
- Percent of County Residents who Identify as White
- Percent of County Residents who Identify as Black
- Percent of County Residents who Identify as a Race Other than White, Black, or Asian

STATISTICAL INFERENCE ON REGRESSION COEFFICIENTS

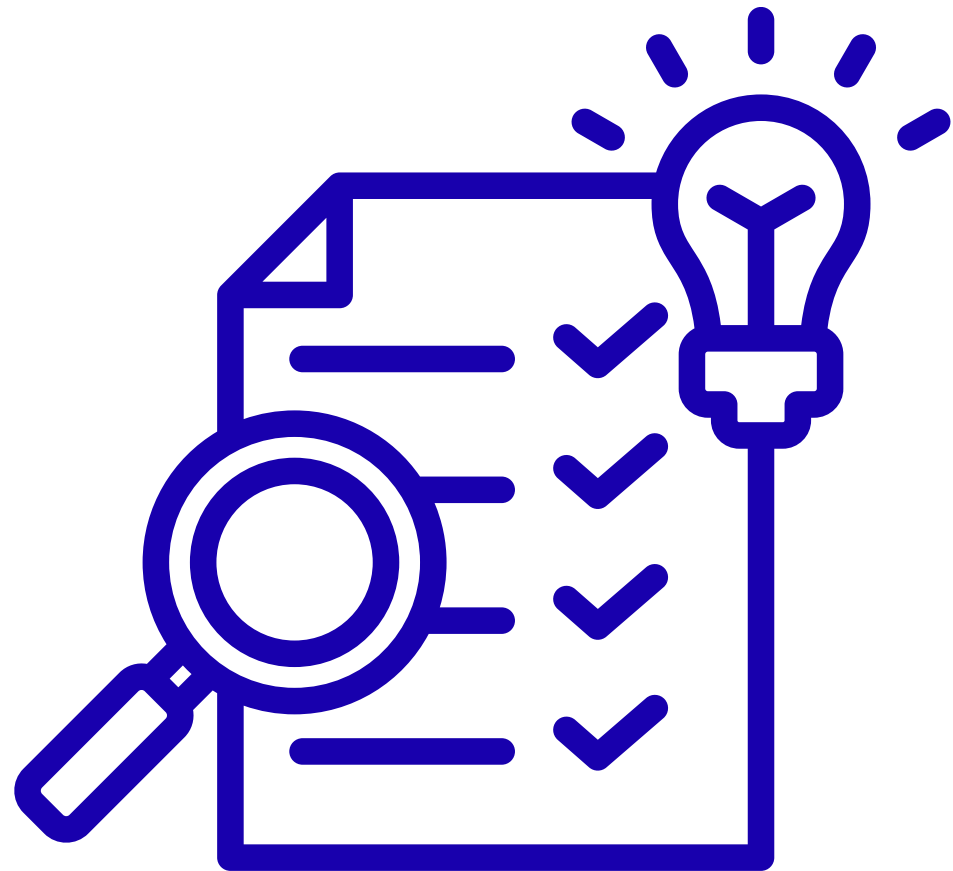
- Statistical Inference on the overall model

H_0 : all $\beta_j = 0$; indicates none of the predictor variables are related to y

H_1 : at least one $\beta_j \neq 0$; indicates at least one of the predictor variables is useful in predicting y or that the model has some ability to predict the variability observed in y

Table 6: Multiple Linear Regression Model Summary with Outliers and Median Age of County Residents
Predictor Variable Removed and ANOVA Test Comparison with Linear Model with Outliers Removed

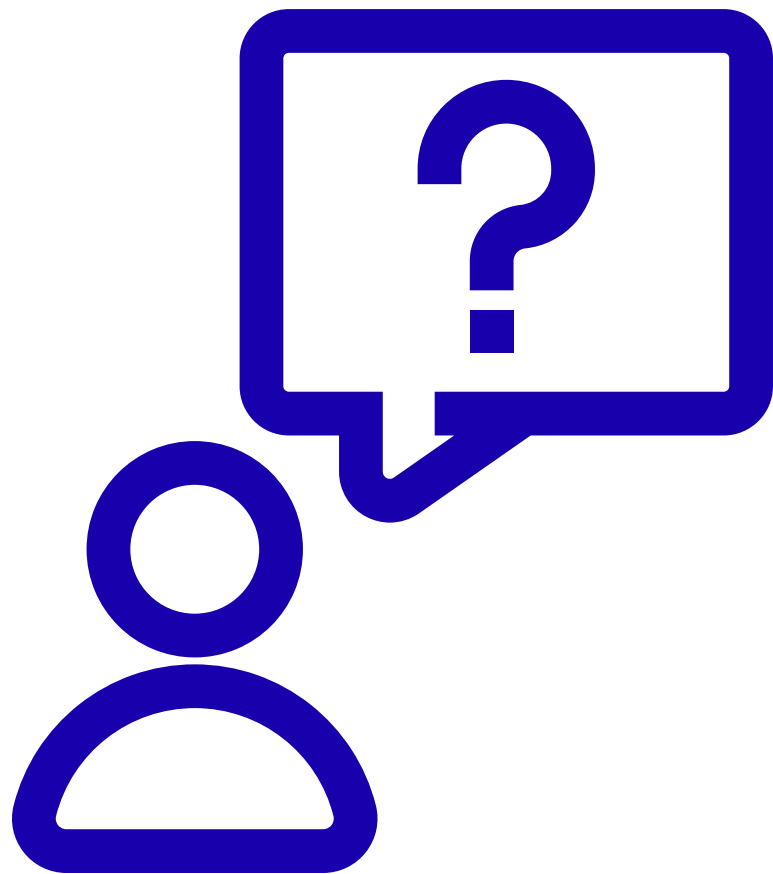
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Regression	0.6189	0.6152	948187	25	37939.5	169.26	<2.2E-16
Error			583951	2606	224.08		
Total			1532138	2631			



CONCLUSION

Socioeconomic factors contribute significantly at the $\alpha = 0.05$ significance level to the prediction model of Cancer Mortality Rates created using the multiple linear regression model method.

QUESTIONS?



REFERENCES

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